#6 (2)

$f(x) = (x+1)^2 \quad (x > -1)$

$x = (y+1)^2$

$\sqrt{x} = y+1$

$\therefore y = \sqrt{x} - 1$

$f^{-1}(x) = \sqrt{x} - 1$

$(f^{-1}(x))' = \frac{1}{2\sqrt{x}}$ //

$(f^{-1}(x))' = \frac{1}{f'(f^{-1}(x))}$

$= \frac{1}{f'(\sqrt{x}-1)}$

$= \frac{1}{2(\sqrt{x}-1+1)}$

$= \frac{1}{2\sqrt{x}}$ ▨

#9. $f(x) = 3x - 1$

$g(x) = \sqrt{2x+5}$ $\quad h(x) = f(g(x)) = 3\sqrt{2x+5} - 1$

$f^{-1}(x) = \frac{1}{3}(x+1)$

$g^{-1}(x) = \frac{1}{2}(x^2-5)$

$h^{-1}(x)$ 구하는 과정 →

$y = 3\sqrt{2x+5} - 1$

$x+1 = 3\sqrt{2y+5}$

$(x+1)^2 = 18y + 45$

$\therefore h^{-1}(x) = \frac{1}{18}(x^2 + 2x - 44)$ ⊥

$g^{-1}(f^{-1}(x)) = \frac{1}{2}\left(\left(\frac{1}{3}x + \frac{1}{3}\right)^2 - 5\right)$

$\therefore h^{-1} = g^{-1} \circ f^{-1}$ ▨